# Worksheet #15

*Week 8, Day 1: Linear regression*

Name: _____
NetID: _____
Date: _____

## Guided checkpoints

For full credit, fill in the checkpoints. If you skip a checkpoint, you must include your own notes for full credit.

### 1. What Is a Model?

Write down a definition of a model, in your own words or from one of the examples on the slides.

### 2. George Box

Write down George Box's quote about "all models."

### 3. Mechanistic or Statistical?

Write down each example and whether it is mechanical or statistical.

A.

B.

C.

D.

### 4. Loss Functions

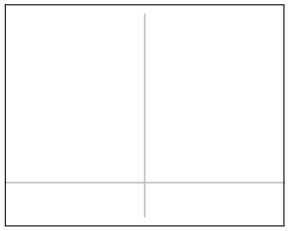
Write the average loss $L(\theta, \mathbf{y})$ from the slides, then explain it in plain language.

Formula:

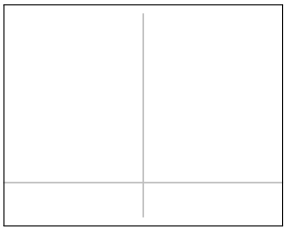
In plain language:

**5. Squared vs. Absolute Error**

Sketch each loss against the error $y - \theta$.



squared error



absolute error

**6. The Modeling Process**

List the main steps of modeling, in order.

1.

2.

3.

4.

**7. Linear Models**

Write the equation for each model.

Constant model:

Simple linear regression:

**8. LINE**

Name each LINE criterion and what it requires.

L:

I:

N:

E:

**9. Multiple Linear Regression**

Write the equation for multiple linear regression.